

Note: Pancyclicity of the Prism

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Abstract

We show that for every 3-connected cubic graph G , the prism $G \times K_2$ has cycles of every even length. Furthermore, if G has a triangle, then $G \times K_2$ is pancyclic.

Key words: Graphs, cubic graphs, prism, cartesian product, pancyclicity

1 Introduction

The prism of a graph G is defined as the cartesian product $G \times K_2$; that is, take two disjoint copies of G and add a matching joining the corresponding vertices in the two copies. Building on earlier work in [1,3,4] and others, Paulraja proved:

Theorem 1 [5] *If G is a 3-connected 3-regular graph, then the prism $G \times K_2$ is hamiltonian.*

In this note we extend Paulraja's result. We show that for every 3-connected 3-regular graph G , the prism $P = G \times K_2$ is "vertex even pancyclic:" for each

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vertex v of P there is in P a cycle of every even length from 4 up to the order of P which contains v . Furthermore, if G has a triangle, then P is pancyclic: it has cycles of every length from 3 up to its order.

2 Proofs

In order to establish our results, we need to review the ideas of the proof of Theorem 1. The key idea is to use a spanning cactus of G .

For graph theory terminology we follow [2]. A vertex of degree 1 is called an *end-vertex*. A *block* is a maximal 2-connected subgraph. A *cactus* is a connected graph every block of which is a K_2 or a cycle. A *3-cactus* is a cactus with maximum degree at most 3 and such that every vertex of degree 3, if any, belongs to a cycle. An *even 3-cactus* is a bipartite 3-cactus (that is, every cycle has even length). A *cubic graph* is a graph in which every vertex has degree 3. A spanning subgraph is one with the same vertex set.

Theorem 2 [5] *If G is a 3-connected cubic graph, then there exists a spanning even 3-cactus H .*

Theorem 3 [5] *If H is an even 3-cactus, then the prism $H \times K_2$ is hamiltonian.*

(We note in passing that Fleischner [3] building on work of Goodey & Rosenfeld [4] showed that the prism of a cubic graph is hamiltonian if and only if the graph contains what he called a BEPS-graph. It can be shown that a minimal BEPS-graph is a spanning even 3-cactus, and so a cubic graph is hamiltonian if and only if it contains a spanning even 3-cactus.)

To prove our results, we need the following observation:

Observation 4 *If H is a 3-cactus of order at least 2, then there exists two vertices v_1 and v_2 such that $H - v_1$ and $H - v_2$ are both 3-acti.*

PROOF. Consider the block graph H' formed by contracting each cycle of H to a single vertex. If H' has one vertex, then H was a cycle, and v_i can be any vertex.

Otherwise, H' contains two end-vertices. Each end-vertex of H' corresponds either to an end-vertex w of H , or to a cycle C in H in which exactly one vertex has degree 3. In the former case, $H - w$ is a 3-cactus; in the latter, $H - x$ is a 3-cactus for any neighbor x of w in C . $\square$

Theorem 5 *If H is an even 3-cactus, then the prism $H \times K_2$ is vertex even pancyclic.*

PROOF. Let v^* be any vertex of $H \times K_2$ and let k be an integer such that $2 \leq k \leq p(H)$. Say v^* corresponds to $v \in V(H)$. By the above observation, H has an even 3-cactus H' of order k containing v . By Theorem 3, $H' \times K_2$ is hamiltonian, and so H has a $2k$ -cycle containing v^* . $\square$

As a direct consequence of Theorems 2 and 5 we get:

Theorem 6 *If G is a 3-connected cubic graph, then the prism $G \times K_2$ is vertex even pancyclic.*

Finally, we show that prisms with triangles are pancyclic:

Theorem 7 *If G is a 3-connected cubic graph that contains a triangle, then the prism $G \times K_2$ is pancyclic.*

PROOF. Suppose that G contains a triangle $T = \{x, y, z\}$. It is clear that $T \times K_2$ contains a 5-cycle and that $K_4 \times K_2$ has a 7-cycle. So assume $p(G) \geq 6$ and we wish to find a $(2k + 1)$ -cycle for $3 \leq k < p(G)$.

Let G' be the graph obtained from G by contracting T to a single vertex t . Assume the two vertices corresponding to t in the prism $G' \times K_2$ are t_1 and t_2 . Since G is 3-connected and $p(G) \geq 6$, the graph G' is 3-connected and cubic. By Theorem 2 and the proof of Theorem 5, G' contains a cycle C' of length $2k - 2$ that contains both t_1 and t_2 .

Now we will convert C' to a cycle of G by replacing $\{t_1, t_2\}$ with five vertices of $T \times K_2$. If C' does not use the edge $t_1 t_2$, then we can replace t_1 by two vertices and t_2 by three vertices and hence this case is handled.

So assume C' uses the edge $t_1 t_2$. Suppose the third neighbours of x, y and z are a, b and c respectively. For each vertex v in G let v_1 and v_2 be the two vertices corresponding to v in $G \times K_2$. Then, without loss of generality we may assume C' contains either the segment a_1, t_1, t_2, a_2 or the segment a_1, t_1, t_2, b_2 . These can be replaced by segments $a_1, x_1, y_1, z_1, z_2, x_2, a_2$ and $a_1, x_1, y_1, z_1, z_2, y_2, b_2$ respectively. This completes the proof. $\square$

It is also possible to show that in general if a cubic graph with a triangle has a spanning even 3-cactus then the resultant prism is pancyclic. For example,

Fleischner [3] showed that a 2-connected planar cubic graph has a BEPS-graph, and therefore is hamiltonian; so, if a 2-connected planar cubic graph has a triangle then its prism is pancyclic. The situation for nonplanar graphs is unresolved.

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